

Version 2

**DEPARTMENT OF THE NAVY (DON)
Small Business Innovation Research (SBIR)
DoW 2026 SBIR CSO Release 5
Phase I Open Topics Proposal Submission Instructions**

IMPORTANT

- **The following instructions apply to topic:**
 - **DON26BX05-NP003**
 - **DON26BX05-NP004**
- Submitting small business concerns (SBCs) are encouraged to thoroughly review the DoW SBIR/STTR Program Commercial Solutions Opening (CSO) and register for the DSIP Listserv to remain apprised of important programmatic changes.
 - The DoW Program CSO is located at: <https://www.dodsbirsttr.mil/submissions/login>.
 - Register for the DSIP Listserv at: <https://www.dodsbirsttr.mil/submissions/login>.
- The information provided in the DON Proposal Submission Instructions takes precedence over the DoW Instructions posted for this CSO.
- Proposing SBCs that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF) or any combination of these are eligible to submit proposals in response to DON topics advertised in this CSO.
- Phase I Technical Volume (Volume 2) and Supporting Documents (Volume 5) templates, specific to DON topics, are available at https://www.navysbir.com/links_forms.htm.
- The DON provides notice that Prototype Other Transaction (OT) will be used for Phase I.
- This CSO is issued under regulations set forth in DFARS Subpart 212.70. and is considered competitive for the purposes of 10 U.S.C. chapter 127 and FAR 6.102. DON will treat items, technologies, and services acquired under this CSO as commercial items. This CSO includes no requirement for a commercial product or service to be an already-developed, off-the-shelf item. Please review the DoW SBIR/STTR Program CSO for more information.

INTRODUCTION

The DON SBIR/STTR Programs are mission-oriented programs that integrate the needs and requirements of the DON's Fleet through research and development (R&D) topics that have dual-use potential, but primarily address the needs of the DON. More information on the programs can be found on the DON SBIR/STTR website at www.navysbir.com.

For questions regarding this CSO use the information in Table 1.

TABLE 1: POINTS OF CONTACT FOR QUESTIONS REGARDING THIS CSO

Type of Question	When	Contact Information
Program and administrative	Always	Navy SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-

Version 2

		sttr@us.navy.mil or appropriate Program Manager listed in Table 2 (below)
Topic-specific technical questions	CSO Pre-release	Technical Point of Contact (TPOC) listed in each topic on the DoW SBIR/STTR Innovation Portal (DSIP). Refer to the Proposal Submission section of the DoW SBIR/STTR Program CSO for details.
	CSO Open	DoW SBIR/STTR Topic Q&A platform (https://www.dodsbirsttr.mil/submissions) Refer to the Proposal Submission section of the DoW SBIR/STTR Program CSO for details.
Electronic submission to the DoW SBIR/STTR Innovation Portal (DSIP)	Always	DSIP Support via email at dodsbirsupport@reisystems.com
Navy-specific CSO instructions and forms	Always	DON SBIR/STTR Program Management Office usn.pentagon.cnr-arlington-va.mbx.navy-sbir-sttr@us.navy.mil

TABLE 2: DON SYSTEMS COMMANDS (SYSCOM) SBIR PROGRAM MANAGERS

Topic Numbers	Point of Contact	SYSCOM	Email
DON26BX05-NP003	Mr. Jason Schroepfer	Naval Sea Systems Command (NAVSEA)	NSSC_SBIR.fct@navy.mil
DON26BX05-NP004	Mr. Shadi Azoum	Naval Information Warfare Systems Command (NAVWAR)	info@navwarsbir.com

OPEN TOPIC SUBMISSIONS:

This CSO release contains Open Topics. As outlined in section 7 of the SBIR and STTR Extension Act of 2022, innovation Open Topic activities—

- a. Increase the transition of commercial technology to the DoW;
- b. Expand the small business nontraditional industrial base;
- c. Increase commercialization derived from investments of the DoW; and
- d. Expand the ability for qualifying SBCs to propose technology solutions to meet the needs of the DoW.

Unlike Conventional Topics, which specify the desired technical objective and output, Open Topics can use generalized mission requirements or specific technology areas to adapt commercial products or solutions to close capability gaps, improve performance, or provide technological advancements in existing capabilities.

An SBC may only submit one (1) proposal to each Open Topic. If more than one proposal from an SBC is received for a single Open Topic, only the most recent proposal to be certified and submitted in DSIP prior to the submission deadline will receive an evaluation. All prior proposals submitted by the SBC for the same Open Topic will be marked as nonresponsive and will not receive an evaluation.

Version 2

PHASE I SUBMISSION REQUIREMENTS

The following section details requirements for submitting a compliant Phase I proposal to the DoW SBIR/STTR Programs.

NOTE: Proposing SBCs are advised that support contract personnel will be used to carry out administrative functions and may have access to proposals, contract award documents, contract deliverables, and reports. All support contract personnel are bound by appropriate non-disclosure agreements.

DoW SBIR/STTR Innovation Portal (DSIP). Proposing SBCs are required to submit proposals via the DoW SBIR/STTR Innovation Portal (DSIP); and follow proposal submission instructions in the DoW SBIR/STTR Program CSO on the DSIP at <https://www.dodsbirsttr.mil/submissions>. Please refer to the DoW SBIR/STTR Program CSO for further information.

Proposal Volumes. The following seven volumes are required.

- **Proposal Cover Sheet (Volume 1).** As specified in DoW SBIR/STTR Program CSO.
- **Technical Proposal (Volume 2)**
 - Technical Proposal (Volume 2) must meet the following requirements or the proposal will be REJECTED:
 - Not to exceed ten (10) pages, regardless of page content
 - Single column format, single-spaced typed lines, 8 ½” x 11” paper, margins one inch on all sides, font size of not less than 10-point
 - Include, within the ten-page limit of Volume 2, an Option that furthers the effort in preparation for Phase II and will bridge the funding gap between the end of Phase I and the start of Phase II. Tasks for both the Phase I Base and the Phase I Option must be clearly identified. Phase I Options are exercised upon selection for Phase II.
 - Work proposed for the Phase I Base must be exactly six (6) months.
 - Work proposed for the Phase I Option must be exactly six (6) months.
 - Additional information:
 - A Phase I Open Topic proposal template specific to DON to meet Phase I requirements is available at https://navysbir.com/links_forms.htm
- **Cost Volume (Volume 3).**
 - Cost Volume (Volume 3) must meet the following requirements or the proposal will be REJECTED:
 - The Phase I Base amount must not exceed \$200,000.
 - Phase I Option amount must not exceed \$115,000.
 - Costs for the Base and Option must be separated and clearly identified in Volume 3.
 - For Phase I, a minimum of two-thirds of the work is performed by the proposing SBC. The two-thirds percentage of work requirement must be met in the Base costs as well as in the Option costs. DON will not accept deviations from the minimum percentage of work requirements for Phase I. The percentage of work is measured by both direct and indirect costs. The following is used to calculate the minimum percentage of work:
 - Proposing SBC Costs (numerator for SBC calculation):
 - Total Firm Costs
 - NOTE: G&A, if proposed, will only be attributed to the proposing SBC.
 - Subcontractor Costs (numerator for subcontractor calculation):

Version 2

- Total Subcontractor Costs (TSC)
 - Total Cost (i.e., Total Cost before Profit Rate is applied, denominator for either calculation)
 - **Cost Sharing: Cost sharing is not accepted on DON Phase I proposals. A value above or below \$0.00 entered in the Cost Sharing field will not be considered in the Phase I contract award.**
- Additional information:
 - Provide sufficient detail for subcontractor, material, and travel costs. Subcontractor costs must be detailed to the same level as the prime contractor. Material costs must include a listing of items and cost per item. Travel costs must include the purpose of the trip, number of trips, location, length of trip, and number of personnel.
 - Inclusion of cost estimates for travel to the sponsoring SYSCOM's facility for one day of meetings is recommended for all proposals.
 - The "Additional Cost Information" of Supporting Documents (Volume 5) may be used to provide supporting cost details for Volume 3. A selected proposal may be required to submit further documentation to the SYSCOM Contracting Officer to substantiate costs.
- **Company Commercialization Report (Volume 4).** DoW collects and uses Volume 4 and DSIP requires Volume 4 for proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoW SBIR/STTR Program CSO for details to ensure compliance with DSIP Volume 4 requirements.
- **Supporting Documents (Volume 5).** Volume 5 is for the submission of administrative material that DON may require to process a proposal, if selected, for contract award.
 - Proposing SBCs must review and submit the following items, as applicable:
 - **Majority Ownership in Part.** Proposing SBCs that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF), or any combination of these as set forth in 13 C.F.R. § 121.702, are eligible to submit proposals in response to DON topics advertised within this CSO. Complete the certification as detailed under ADDITIONAL SUBMISSION CONSIDERATIONS.
 - Additional information:
 - Proposing SBCs may include administrative materials in Supporting Documents (Volume 5); a template is available at https://navysbir.com/links_forms.htm to provide guidance on optional material the proposing SBC may include in Volume 5:
 - Additional Cost Information to support the Cost Volume (Volume 3)
 - SBIR/STTR Funding Agreement Certification
 - Data Rights Assertion
 - Disclosure of Information (DFARS 252.204-7000)
 - Prior, Current, or Pending Support of Similar Proposals or Awards
 - Foreign Citizens
 - Details of Request for Discretionary Technical and Business Assistance (TABAs), if proposed, is to be included under the Additional Cost Information section if using the DON Supporting Documents template.
 - Do not include documents or information to substantiate the Technical Volume (Volume 2) in Volume 5 (e.g., resumes, test data, technical reports, or publications). Such documents or information will not be considered.

Version 2

- **Fraud, Waste and Abuse Training (Volume 6).** DoW requires Volume 6 for submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoW SBIR/STTR Program CSO for details.
- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7).** In accordance with 15 U.S.C. §638(vv) and the SBA SBIR/STTR Policy Directive, the DoW will review all proposals submitted in response to this CSO to assess security risks presented by SBCs seeking a Federally funded award. SBCs must complete the Disclosures of Foreign Affiliations or Relationships to Foreign Countries webform in Volume 7 of the DSIP proposal submission. Please refer to the Proposal Preparation Instructions and Requirements section of the DoW SBIR/STTR Program CSO for details.

PHASE I EVALUATION AND SELECTION

The following section details how the DON SBIR/STTR Programs will evaluate Phase I proposals.

Proposals meeting DSIP submission requirements will be forwarded to the DON SBIR/STTR Programs. Prior to evaluation, all proposals will undergo a compliance review to verify compliance with DoW and DON SBIR/STTR proposal eligibility requirements. Proposals not meeting submission requirements will be REJECTED and not evaluated. Due to limited funding, the DON reserves the right to limit the number of awards under any topic.

- **Proposal Cover Sheet (Volume 1).** The Proposal Cover Sheet (Volume 1) will undergo a compliance review to verify the proposing SBC has met eligibility requirements and followed the instructions for the Proposal Cover Sheet as specified in the DoW SBIR/STTR Program CSO.
- **Technical Volume (Volume 2).** The DON will evaluate and select Phase I proposals using the evaluation criteria specified in the Method of Selection and Evaluation Criteria section of the DoW SBIR/STTR Program CSO, with technical merit being most important, followed by qualifications of key personnel and commercialization potential of equal importance. The information considered for this decision will come from Volume 2. This is not a FAR Part 15 evaluation. Proposals will not be compared to one another.
- **Cost Volume (Volume 3).** The Cost Volume (Volume 3) will not be considered in the selection process and will only undergo a compliance review to verify the proposing SBC has met the requirements of Volume 3.
- **Company Commercialization Report (CCR) (Volume 4).** Not evaluated.
- **Supporting Documents (Volume 5).** Supporting Documents (Volume 5) will not be considered in the selection process and will only undergo a compliance review to ensure the proposing SBC has included items in accordance with the PHASE I SUBMISSION REQUIREMENTS section above.
- **Fraud, Waste, and Abuse Training (Volume 6).** Not evaluated.
- **Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7).** Disclosures of Foreign Affiliations or Relationships to Foreign Countries (Volume 7) will be assessed as part of the Foreign Risk Evaluation (FRE) Program to Assess Security Risks. Refer to the DoW SBIR/STTR Program CSO to ensure compliance with Volume 7 requirements.

Version 2

ADDITIONAL SUBMISSION CONSIDERATIONS

This section details additional items for proposing SBCs to consider during proposal preparation and submission process.

Foreign Risk Evaluation (FRE) Program to Assess Security Risks. 15 U.S.C. §638(vv) requires the Department of War, in coordination with the Small Business Administration, to establish and implement a due diligence program, known as Foreign Risk Evaluation (FRE) Program, to assess and, when possible, mitigate security risks presented by SBCs seeking a Federally-funded award. Please review the Certifications and Registrations section of the DoW SBIR/STTR Program CSO for details on how DoW will assess security risks presented by SBCs. The FRE Program to Assess Security Risks will be implemented for all Phases.

Discretionary Technical and Business Assistance (TABA). The Small Business Innovation and Economic Security Act Section 7 mandates agencies offer TABA. The purpose of TABA is to assist awardees in making better technical decisions on SBIR/STTR projects; solving technical problems that arise during SBIR/STTR projects; minimizing technical risks associated with SBIR/STTR projects; commercializing the SBIR/STTR product or process, including intellectual property protections and cybersecurity assistance; and screening for potential foreign involvement in technology development or commercialization activities. TABA services can be provided through vendor(s) selected by the SBC or the SBC may propose use of TABA funding to hire new staff, augment staff, or direct staff to conduct or participate in training activities consistent with the purpose of TABA as detailed above. The Phase I TABA amount, if proposed, cannot include any profit/fee by the proposing SBC and must be inclusive of all applicable indirect costs. TABA cannot be used in the calculation of general and administrative expenses (G&A) for the SBIR proposing SBC. An SBC receiving TABA will be required to submit a report detailing the results and benefits of the service received. This TABA report will be due at the time of submission of the final report.

Request for TABA funding will be reviewed by the DON SBIR/STTR Program Management Office.

If proposing a TABA Provider, the following MUST be included in the request or be subject to denial.

- Provider(s)
- Provider(s) point of contact, email address, and phone number
- An explanation of the provider's unique qualifications to provide the TABA service
- Tasks that will be performed by the provider (to include the purpose and objective of the assistance)
- Total provider(s) cost, number of hours, and labor rates (average/blended rate is acceptable)

If proposing TABA funding to hire new staff, augment staff, or direct staff to conduct or participate in training activities consistent with the purpose of TABA, the following MUST be included or be subject to denial.

- Name(s), position(s), business need to be filled, and/or training to be provided
- Number of employees to be hired, augmented, or directed to participate in training activities
- Qualifications of employees hired or augmented, or detailed need for training
- Tasks that will be performed (to include the purpose and objective of the assistance)
- Total staff/training cost, number of hours, and labor rates (average/blended rate is acceptable)

TABA requests must be included in the proposal as follows:

- Phase I:
 - Online DoW Cost Volume (Volume 3) – the value of the TABA request.

Version 2

- Supporting Documents (Volume 5) – a detailed request for TABA (as specified above) specifically identified as “TABA” in the section titled Additional Cost Information when using the DON Supporting Documents template.
- Phase II:
 - DON Phase II Cost Volume (provided by the DON SYSCOM) - the value of the TABA request.
 - Supporting Documents (Volume 5) – a detailed request for TABA (as specified above) specifically identified as “TABA” in the section titled Additional Cost Information when using the DON Supporting Documents template.

Proposed values for TABA must NOT exceed:

- Phase I: A total of \$6,500, in addition to the Phase I award amount.
- Phase II: A total of \$25,000 per award, not to exceed \$50,000 per Phase II project, included in the Phase II award amount.

Disclosure of Information (DFARS 252.204-7000). In order to eliminate the requirements for prior approval of public disclosure of information (in accordance with DFARS 252.204-7000) under this award, the proposing SBC shall identify and describe all fundamental research to be performed under its proposal, including subcontracted work, with sufficient specificity to demonstrate that the work qualifies as fundamental research. Fundamental research means basic and applied research in science and engineering, the results of which ordinarily are published and shared broadly within the scientific community, as distinguished from proprietary research and from industrial development, design, production, and product utilization, the results of which ordinarily are restricted for proprietary or national security reasons (defined by National Security Decision Directive 189). An SBC whose proposed work will include fundamental research and requests to eliminate the requirement for prior approval of public disclosure of information must complete the DON Fundamental Research Disclosure and upload as a separate PDF file to the Supporting Documents (Volume 5) in DSIP as part of their proposal submission. The DON Fundamental Research Disclosure is available on https://navysbir.com/links_forms.htm and includes instructions on how to complete and upload the completed Disclosure. Simply identifying fundamental research in the Disclosure does **NOT** constitute acceptance of the exclusion. All exclusions will be reviewed and, if approved by the Government Contracting Officer, noted in the contract.

Majority Ownership in Part. Proposing SBCs that are more than 50% owned by multiple venture capital operating companies (VCOC), hedge funds (HF), private equity firms (PEF), or any combination of these as set forth in 13 C.F.R. § 121.702, **are eligible** to submit proposals in response to DON topics advertised within this CSO.

For proposing SBCs that are a member of this ownership class the following must be satisfied for proposals to be accepted and evaluated:

- a. Prior to submitting a proposal, SBCs must register with the SBA Company Registry Database.
- b. The proposing SBC within its submission must submit the Majority-Owned VCOC, HF, and PEF Certification. A copy of the SBIR VC Certification can be found on https://navysbir.com/links_forms.htm. Include the SBIR VC Certification in the Supporting Documents (Volume 5).
- c. Should a proposing SBC become a member of this ownership class after submitting its proposal and prior to any receipt of a funding agreement, the proposing SBC must immediately notify the Contracting Officer, register in the appropriate SBA database, and submit the required certification, which can be found on https://navysbir.com/links_forms.htm.

System for Award Management (SAM). It is strongly encouraged that proposing SBCs register in SAM, <https://sam.gov>, by the Close date of this CSO, or verify their registrations are still active and will

Version 2

not expire within 60 days of CSO Close. Additionally, proposing SBCs should confirm that they are registered to receive contracts (not just grants) and the address in SAM matches the address on the proposal. An SBC selected for an award MUST have an active SAM registration at the time of award or they will be considered ineligible.

Cybersecurity Maturity Model Certification (CMMC) Program. DoW has established the CMMC Program to verify that awardees have implemented required security measures necessary to safeguard Federal Contract Information (FCI) and Controlled Unclassified Information (CUI). CMMC Level requirements are identified within each topic and due at time of award. Proposing SBCs should anticipate that a Projected CMMC Level for Phase II award may be higher than the Projected CMMC Level advertised in the Phase I topic. Proposing SBCs should carefully review and consider the CMMC requirements as compliance may impact proposed costs and technical approach. Please review the DoW SBIR/STTR Program BAA for additional information on the CMMC Program.

Notice of NIST SP 800-171 Assessment Database Requirement. The purpose of the National Institute of Standards and Technology (NIST) Special Publication (SP) 800-171 is to protect Controlled Unclassified Information (CUI) in Nonfederal Systems and Organizations. As prescribed by DFARS 252.240-7997, in order to be considered for award, an SBC is required to implement NIST SP 800-171 and shall have a current assessment uploaded to the Supplier Performance Risk System (SPRS) which provides storage and retrieval capabilities for this assessment. The platform Procurement Integrated Enterprise Environment (PIEE) will be used for secure login and verification to access SPRS. For brief instructions on NIST SP 800-171 assessment, SPRS, and PIEE, please visit <https://www.sprs.csd.disa.mil/nistsp.htm>. For in-depth tutorials on these items, please visit <https://www.sprs.csd.disa.mil/webtrain.htm>.

Human Subjects, Animal Testing, and Recombinant DNA. Due to the short timeframe associated with Phase I of the SBIR/STTR process, the DON does **not** recommend the submission of Phase I proposals that require the use of Human Subjects, Animal Testing, or Recombinant DNA. The ability to obtain Institutional Review Board (IRB) approval for proposals that involve human subjects can take 6-12 months, and that lengthy process can be at odds with the Phase I goal for time-to-award. Before the DON makes any award that involves an IRB or similar approval requirement, the proposing SBC must demonstrate compliance with relevant regulatory approval requirements that pertain to proposals involving human, animal, or recombinant DNA protocols. It will not impact the DON's evaluation, but requiring IRB approval may delay the start time of the Phase I award and if approvals are not obtained within two months of notification of selection, the decision to award may be terminated. If the use of human, animal, and recombinant DNA is included under a Phase I or Phase II proposal, please carefully review the requirements at: <https://www.nre.navy.mil/work-with-us/how-to-apply/compliance-and-protections/research-protections>. This webpage provides guidance and lists approvals that may be required before contract/work can begin.

Government Furnished Equipment (GFE). Due to the typical lengthy time for approval to obtain GFE, it is recommended that GFE is not proposed as part of the Phase I proposal. If GFE is proposed, and it is determined during the proposal evaluation process to be unavailable, proposed GFE may be considered a weakness in the technical merit of the proposal.

International Traffic in Arms Regulation (ITAR). For topics indicating ITAR restrictions or the potential for classified work, limitations are generally placed on disclosure of information involving topics of a classified nature or those involving export control restrictions, which may curtail or preclude the involvement of universities and certain non-profit institutions beyond the basic research level. SBCs must structure their proposals to clearly identify the work that will be performed that is of a basic research nature and how it can be segregated from work that falls under the classification and export control

Version 2

restrictions. As a result, information must also be provided on how efforts can be performed in later phases if the university/research institution is the source of critical knowledge, effort, or infrastructure (facilities and equipment).

SELECTION, AWARD, AND POST-AWARD INFORMATION

Notifications. Email notifications for proposal receipt (approximately one week after the Phase I CSO Close) and selection are sent based on the information received on the proposal Cover Sheet (Volume 1). Consequently, the e-mail address on the proposal Cover Sheet must be correct.

Proposal feedback. DON anticipates fulfilling proposal feedback requests using automated standardized language associated with the technical evaluation. Proposal feedback is not the same as a FAR Part 15 debriefing.

Protests. Interested parties have the right to protest in accordance with the procedures in FAR Subpart 33.1.

Protests related to the terms of the CSO must be served to: osd.ncr.ousd-r-e.mbx.SBIR-STTR-Protest@mail.mil. A copy of a pre-award Government Accountability Office (GAO) protest must also be filed with the aforementioned email address within one day of filing with the GAO.

Protests related to a selection or award decision should be filed with the appropriate Contracting Officer for an Agency Level Protest or with the GAO. Contracting Officer contact information for specific DON Topics may be obtained from the DON SYSCOM Program Managers listed in Table 2 above. For protests filed with the GAO, a copy of the protest must be submitted to the appropriate DON SYSCOM Program Manager and the appropriate Contracting Officer within one day of filing with the GAO.

Awards. Due to limited funding, the DON reserves the right to limit the number of awards under any topic. Any notification received from the DON that indicates the proposal has been selected does not ultimately guarantee an award will be made. This notification indicates that the proposal has been selected in accordance with the evaluation criteria and has been sent to the Contracting Officer to conduct compliance review to confirm eligibility of the proposing SBC, and to take other relevant steps necessary prior to making an award.

Award Type. Phase I awards to this CSO will be executed as Prototype Other Transaction (OT) under 10 U.S.C. 4021/10 U.S.C. 4022.

Funding Limitations. The maximum Phase I proposal/award amount including all options is \$315,000. The Phase I Base amount must not exceed \$200,000 and the Phase I Option amount must not exceed \$115,000

Deliverables. Deliverables for Phase I are typically a kick-off brief, progress reports, and a final report. Required contract deliverables (as stated in the contract) must be uploaded to <https://www.navysbirprogram.com/navydeliverables/>.

Payments. The DON will make **three** payments from the start of the Open Topic Phase I Base period, and three payments from the start of the Open Topic Phase I Option period, if exercised. Payment amounts represent a set percentage of the Base or Option value as follows:

Days From Start of Base	Payment Amount
15 Days	50% of Total Base
90 Days	35% of Total Base

Version 2

180 Days	15% of Total Base
Days From Start of Option	Payment Amount
15 Days	50% of Total Option
90 Days	35% of Total Option
180 Days	15% of Total Option

PHASE II GUIDELINES

Evaluation and Selection. All Phase I awardees may submit an **Initial** Phase II proposal for evaluation and selection. The evaluation criteria for Phase II is the same as Phase I (as stated in this CSO). The Phase I Final Report and Initial Phase II Proposal will be used to evaluate the SBC's potential to progress to a workable prototype in Phase II and transition the technology to Phase III. Details on the due date, content, and submission requirements of the Initial Phase II Proposal will be provided by the awarding SYSCOM either in the Phase I contract or by subsequent notification.

Awards. Awards to this CSO will be executed as Prototype Other Transaction (OT) under 10 U.S.C. 4021/10 U.S.C. 4022. Phase II awards can be structured in a way that allows for increased funding levels based on the project's transition potential.

Navy SBIR Transition Program (STP). All Phase II awardees have the opportunity to participate in the virtual Navy STP Kickoff during the first or second year of the Phase II contract. While there are no travel costs associated with this virtual event, Phase II awardees should budget time of up to a full day to participate. STP information can be obtained at: <https://navystp.com>. Phase II awardees will be contacted separately regarding this program.

Funding Limitations. The maximum Phase II proposal/award amount including all options (including TABA) is \$2,000,000 (unless non-SBIR/STTR funding is being added). Individual SYSCOMs may award amounts, including Base and all Options, of less than \$2,000,000 based on available funding. The structure of the Phase II proposal/award will be provided to all Phase I awardees. In accordance with the SBIR and STTR Policy Directive section 4(b)(5), there is a limit of one sequential Phase II award per SBC per topic.

PHASE III GUIDELINES

A Phase III SBIR/STTR award is any work that derives from, extends, or completes effort(s) performed under prior SBIR/STTR funding agreements, but is funded by sources other than the SBIR/STTR programs. This covers any contract, grant, or agreement issued as a follow-on Phase III award or any contract, grant, or agreement award issued as a result of a competitive process where the awardee was an SBIR/STTR firm that developed the technology as a result of a Phase I or Phase II award. The DON will give Phase III status to any award that falls within the above-mentioned description. Consequently, DON will assign SBIR/STTR Data Rights to any noncommercial technical data and noncommercial computer software delivered in Phase III that were developed under SBIR/STTR Phase I/II effort(s). Government prime contractors and their subcontractors must follow the same guidelines as above and ensure that companies operating on behalf of the DON protect the rights of the SBIR/STTR firm.

Version 2

NAVY SBIR DoW 2026 CSO

Topic Index

Release 5

DON26BX05-NP003	NAVSEA Open Topic for Model-Based Systems Engineering (MBSE) Acceleration via Artificial Intelligence or Machine Learning Data Ingestion
DON26BX05-NP004	NAVWAR Open Topic for Unified Assured Positioning, Navigation, and Timing Operational Awareness and Decision Support

Version 2

DON26BX05-NP003 TITLE: NAVSEA Open Topic for Model-Based Systems Engineering (MBSE)
Acceleration via Artificial Intelligence or Machine Learning Data Ingestion

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Integrated Sensing and Cyber

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

OBJECTIVE: Develop an Artificial Intelligence/Machine Learning (AI/ML) capability to rapidly transition legacy programs to model-based systems engineering (MBSE).

DESCRIPTION: The Navy's AEGIS Combat System was initially developed in the late 1970s and continues in use today, with updates and changes to the core 1970s design. As a result of this multi-year development, much of its system design documentation is only available on paper or in disparate legacy digital formats that are not easily converted to models. Given the continuous development nature of weapons systems, a shift in program focus to modeling would necessarily divert funding away from capability development, critical to meet warfighter needs. As a result, the AEGIS program requires the ability to rapidly and affordably ingest and develop highly accurate system models in disparate formats. The use of model-based systems engineering (MBSE) will enhance the program's ability to provide functional kinetic capabilities and fewer errors to combat emerging cyber and kinetic threats. A model that can use AI/ML to ingest multiple forms of technical documentation (e.g., digital files, images, Adobe files, written documents, installation and interface configurations, software and hardware bills of material, cable run sheets, IP addresses, ports protocols and services, etc.) and auto generate Systems Modeling Language (SYSML) models sufficient to support a system will be beneficial to all types of Navy systems. There is nothing commercially available that does this.

The solution will be capable of supporting system design, capability development, cyber assessment and authorization, interface design and verification, functional testing, configuration management, reliability calculation and tracking, problem tracking and program certification functions. These solution models shall be sufficient to support cyber assessments and digital engineering capabilities needed for rapidly developing and automated security assessment with minimal investment in manual labor for model validation and verification. It will use AI/ML to digitize, integrate, and ingest system technical documentation and automate development of models with no human input and a high degree of accuracy. The solution must provide for direct conversion to MBSE Models in the CAMEO Magic Draw SYSML modeling environment. Solutions should incorporate how data will be converted and integrated into SYSML models. Finished models may require some human modification to accurately represent system design and functionality but should work seamlessly with Cameo system modeler Magic Draw in SYSML.

Work produced in Phase II may become classified. Note: The prospective contractor(s) must be U.S. owned and operated with no foreign influence as defined by 32 U.S.C. § 2004.20 et seq., National

Version 2

Industrial Security Program Executive Agent and Operating Manual, unless acceptable mitigating procedures can and have been implemented and approved by the Defense Counterintelligence and Security Agency (DCSA) formerly Defense Security Service (DSS). The selected contractor must be able to acquire and maintain a secret level facility and Personnel Security Clearances. This will allow contractor personnel to perform on advanced phases of this project as set forth by DCSA and NAVSEA in order to gain access to classified information pertaining to the national defense of the United States and its allies; this will be an inherent requirement. The selected company will be required to safeguard classified material during the advanced phases of this contract IAW the National Industrial Security Program Operating Manual (NISPOM), which can be found at Title 32, Part 2004.20 of the Code of Federal Regulations.

PHASE I: Develop a concept for development tools to rapidly transition legacy programs to MBSE and demonstrate they meet the parameters of the Description. Assess feasibility and estimate initial performance to support the test and operational environments. Feasibility will be established through analysis and modelling. The Phase I Option, if exercised, will include initial design and interface specifications necessary to build and demonstrate the prototype in Phase II.

PHASE II: Develop and deliver a prototype that rapidly transitions legacy programs to MBSE based on the results of Phase I. Demonstrate functionality and performance for ingesting disparate data and providing for rapid conversion to MBSE Models. The application will be implemented in a provided Government-approved modeling and simulation environment such as the Navy's Integrated Modeling Environment, to validate performance. It will be evaluated by Government subject matter experts for validation.

It is probable that the work under this effort will be classified under Phase II (see Description section for details).

PHASE III DUAL USE APPLICATIONS: Support the Navy in transitioning the prototype MBSE development tools to allow for further experimentation and refinement. The prototype application will be incorporated using AEGIS program data. This will consist of integration into a baseline definition, incorporation of the baselines existing and new threat capabilities, validation testing, and combat system certification. Scale and operationalize the use of tools to develop models for other shipboard systems across the PEO IWS, combat system, and C4I elements. Provide opportunities to develop and operationalize model functionality to support programmatic functions such as configuration management, reliability, sustainment, etc. and demonstrate efficiencies in these areas.

The developed technology should be broadly applicable to all types of technical data from installation drawings, cable run sheets, hardware lists, software lists, ACAS scans, technical manuals, digital e-mass records, configuration reports, Dynamic Object-Oriented Requirements System (DOORS), reliability reports, and data used in design and sustainment. Dual use applications are numerous; almost any analyst seeking to combine/utilize MBSE for design or sustainment could use this technology.

REFERENCES:

1. McDermott, Tom; DeLaurentis, Dan; Beling, Peter; Blackburn, Mark and Bone, Mary. "AI4SE and SE4AI: A Research Roadmap." Insight 23(1), March 2020, pp.8-14.
https://www.researchgate.net/publication/340649785_AI4SE_and_SE4AI_A_Research_Roadmap
2. Evans, Dr. Eric D., Chair Defense Science Board. "Digital Engineering Capability to Automate Testing and Evaluation: Final Report May 2024." Office of the Secretary of Defense for Research and Engineering. https://dsb.cto.mil/wp-content/uploads/2024/08/DSB_DE_Final-Report_050124_Stamped.pdf

Version 2

3. National Industrial Security Program Executive Agent and Operating Manual (NISP), 32 U.S.C. § 2004.20 et seq. (1993). <https://www.ecfr.gov/current/title-32/subtitle-B/chapter-XX/part-2004>

KEYWORDS: Model-based systems engineering; MBSE; data conversion; data scanning; automated intelligence; machine learning for documentation conversion; model integration, cyber assessment, and system design

Version 2

DON26BX05-NP004 TITLE: NAVWAR Open Topic for Unified Assured Positioning, Navigation, and Timing Operational Awareness and Decision Support

OUSW (R&E) CRITICAL TECHNOLOGY AREA(S): Applied Artificial Intelligence (AAI)

COMPONENT TECHNOLOGY PRIORITY AREA(S): Human-Machine Interfaces; Integrated Network Systems-of-Systems; Integrated Sensing and Cyber

PROJECTED CMMC LEVEL REQUIREMENT: Level 2 (Self)

OBJECTIVE: Develop and/or adapt innovative software-enabled capabilities that improve operator understanding of Assured Positioning, Navigation, and Timing (APNT) status, confidence, threats, and recovery options across diverse positioning, navigation, timing, and situational awareness systems and providers.

DESCRIPTION: The United States Navy relies on accurate positioning, navigation, and timing information to support maritime operations, distributed forces, mission planning, and operational decision-making. As the Navy continues to field a growing number of APNT capabilities and related situational awareness systems, operators are increasingly required to interact with multiple interfaces and data sources to understand system health, confidence levels, operational risks, and available response options. This fragmented environment increases cognitive workload, complicates training, and slows decision-making.

The Navy's GPS-Based Positioning, Navigation, and Timing Service (GPNTS) program continues to enhance APNT resilience through the development and integration of a diverse set of technologies and systems. Examples of these data-types include the opensource All-Source Positioning and Navigation (ASPN) and PNT operating system (pntOS). The Navy seeks innovative approaches capable of ingesting, integrating, processing, and presenting information through GPNTS, to include alternate PNT sources, and related situational awareness systems through a unified operational experience. Of particular interest are solutions that provide operators with rapid understanding of APNT system status, confidence levels, threats, degradations, operational impacts, and recommended courses of action. A common operating picture, "single pane of glass," decision-support environment, or similar approach may be employed to achieve these objectives. The solutions are to transform fragmented operational data into actionable decision advantage and improve the ability of operators to rapidly sense, understand, and act in contested environments.

This topic does not seek development of new Positioning, Navigation, and Timing (PNT) or APNT sensing technologies, navigation algorithms, timing sources, or hardware systems. Instead, the Department seeks expertise in software architecture, data integration, information management, user experience (UX), human-machine interface (HMI) design, analytics, visualization, and decision-support technologies that improve operator effectiveness. Industry best practices in modern software architecture standards, such as containerization for on-prem isolated runtimes will enable the rapid and secure deployment of this capability while minimizing cybersecurity risks from cross-application dependencies.

Commercial industries such as autonomous transportation, cloud operations, telecommunications, logistics, and industrial automation face similar challenges in integrating information from multiple distributed systems to support rapid operational decision-making. Mature commercial technologies that have demonstrated success in aggregating, correlating, visualizing, and presenting actionable insights from disparate data sources may also be suggested. These may include industrial monitoring systems, transportation management systems, enterprise operations centers, autonomous system operations, telecommunications network management, logistics management platforms, and other environments

Version 2

requiring real-time awareness across disaggregated systems. Solutions do not need to be purpose-built for APNT applications if they can be adapted to address the operational objectives of this topic.

Primary technology areas of interest include, but are not limited to:

- Data integration and orchestration platforms
- Human-machine interface (HMI) design
- User experience (UX) design
- Common operating picture technologies
- Operational data visualization
- Decision-support systems
- Workflow automation
- Human-centered design methodologies
- Data analytics and correlation platforms
- AI-enabled operator assistance
- Containerized software architectures
- Mission management software
- Digital engineering environments

Solutions outside these areas that address the stated objective may also be submitted.

PHASE I: The DON is planning to issue multiple Phase I awards for this topic but reserves the right to issue no awards.

Establish feasibility of the conceptual technology or approach through relevant operational use cases, anticipated integration approach, expected improvements to operator effectiveness, and the path toward implementation within Navy operational environments.

Document the results of Phase I in a Final Report. Prepare a Phase II plan.

Phase I Base Deliverables

- Kick-Off Briefing
- Progress Report
- Final Report
- Initial Phase II Proposal

PHASE II: All Phase I awardees may submit an Initial Phase II proposal for evaluation and selection.

Develop a working prototype that demonstrates measurable improvements in operator situational awareness, information accessibility, decision-making effectiveness, workflow efficiency, confidence assessment, or overall operator effectiveness. Mature, integrate, demonstrate, and validate the proposed capability within a representative operational environment.

(Note: Prototype demonstrations may include integration with representative GPNTS, APNT, situational awareness, or related operational data sources.)

Prepare a final report that includes a transition strategy, deployment considerations, sustainment approach, and commercialization plan.

PHASE III DUAL USE APPLICATIONS: Support transition of the developed capability to Fleet operators and associated Programs of Record.

Version 2

Commercial organizations increasingly require the ability to integrate, organize, visualize, and act upon information from numerous distributed systems. Technologies developed under this topic may have broad applicability in autonomous transportation systems, commercial shipping and logistics, aviation operations, industrial automation, telecommunications network operations, smart infrastructure management, cloud and data center operations, critical infrastructure monitoring, public safety, and enterprise operations centers.

REFERENCES:

1. U.S. Government Accountability Office. "Defense Navigation Capabilities: DOD is Developing Positioning, Navigation, and Timing Technologies to Complement GPS." GAO-21-320SP; Washington, DC, 2021. <https://www.gao.gov/assets/720/714196.pdf>
2. Department of the Navy. "Positioning, Navigation, and Timing (PNT); OPNAVINST 9420.1." Office of the Chief of Naval Operations: Washington, DC, 2019. <https://www.secnav.navy.mil/doni/Directives/09000%20General%20Ship%20Design%20and%20Support/09-400%20Command%20and%20Surveillance%20Systems%20Support/9420.1C.pdf>
3. Integrated Solutions for Systems (IS4S) Team. "ASPN and ntOS: An Open-Source Ecosystem for Building Navigation Systems." Proceedings of the International Technical Meeting of the Institute of Navigation: ION GNSS+, Denver, CO, pp 1121-1132, 2023. <https://www.ion.org/publications/abstract.cfm?articleID=17172>
4. U.S., Department of Defense. "Summary of the Joint All-Domain Command and Control Strategy." U.S. Department of Defense, Mar. 2022, media.defense.gov/2022/Mar/17/2002958406/-1/-1/1/SUMMARY-OF-THE-JOINT-ALL-DOMAIN-COMMAND-AND-CONTROL-STRATEGY.pdf.

KEYWORDS: Assured Positioning, Navigation, and Timing; APNT; GPS-Based Positioning, Navigation, and Timing Service (GPNTS); Alternate PNT; Common Operating Picture; Human-Machine Interface (HMI); User Experience (UX); Data Integration; Data Visualization; Decision Support; Operational Workflows; Information Architecture; Data Analytics; Situational Awareness; Mission Management; Software Architecture